DAY 13 — Time Series Basics & Understanding Stock Market Data

Day 13 was one of the most important days in the entire training program because it introduced the concepts behind **Time Series Analysis**, which served as the foundation for the Tesla stock forecasting project. Time series data behaves very differently compared to traditional datasets, and today's learning helped me understand those differences clearly.

🔹 1. What is a Time Series?

Time series is a sequence of data points recorded over time.

Key characteristics:

- Data is ordered (date/time-dependent)
- Values depend on previous values
- Trends and patterns evolve over time

Examples:

- Stock prices
- Weather data
- Traffic volume
- Temperature readings

This made it easier to see why stock forecasting cannot use normal regression models — it requires time-aware algorithms.

🔸 2. Components of Time Series

We studied four main components:

✓ Trend

Long-term movement (upwards or downwards)

✓ Seasonality

Repeated pattern (weekly, monthly, yearly)

✓ Cyclic Patterns

Irregular ups and downs due to events

✓ Random Noise

Unpredictable fluctuations

Understanding these components is essential to choose the right forecasting model like **Prophet**.

🔸 3. Stock Market Time Series Behavior

Tesla stock data showed:

- High volatility
- Sharp spikes during announcements
- Weekly trading patterns
- Clear upward momentum in 2020

We learned that financial datasets:

- Are highly non-linear
- Contain market shocks
- Require strong uncertainty estimation

🔸 4. Stationarity in Time Series

A stationary series has constant:

- Mean
- Variance
- Covariance

Real stock data is **not** stationary.
Therefore:

- Differencing
- Smoothing
- Log transformation
  help remove trends and stabilize variance.

🔸 5. Visualizing Time Series

We used:

plt.plot()

and Plotly charts to visualize:

- Daily close values

- Moving averages
- Volume trends

These visualizations revealed hidden patterns.

Key Takeaways from Day 13

- Time-series data needs special treatment
- Trends + seasonality are critical for forecasting
- Stock data behaves differently from normal ML datasets
- Prophet handles most time-series complexities automatically